AI Engineering
as a practice

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About Me



- Member of the AI Innovation Hub (AIH) team in Dell Technologies
- 20+ years as an IT Practitioner
- Software Development, Project Management, Operations Management, DevOps, Pre-Sales and Post-Sales
- Passionate about Cloud Native technologies and their emerging trends

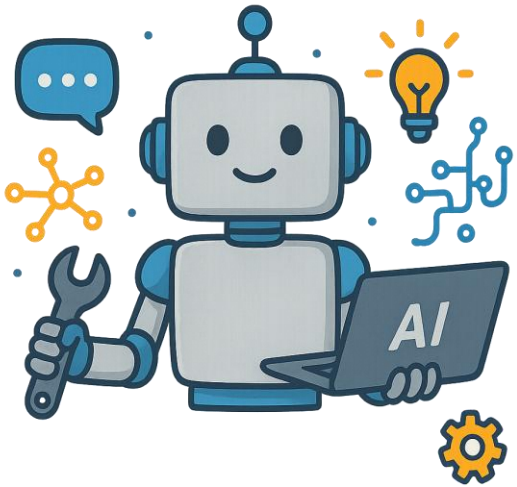
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What is AI Engineering?

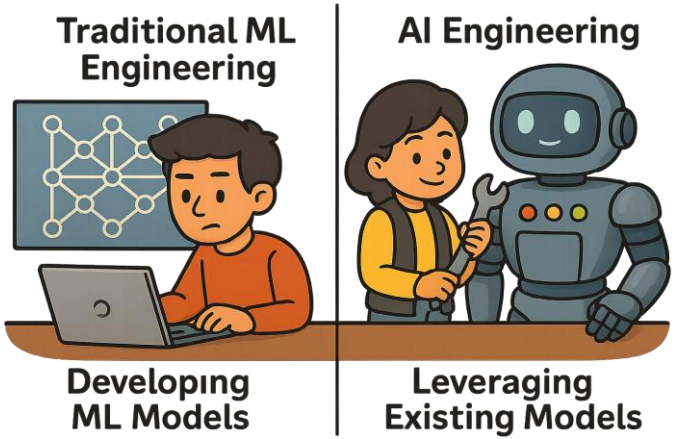
- A discipline that applies engineering principles to *design*, *develop*, and *deploy* AI systems
- A process of building applications on top of **existing** foundation models



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Moving from traditional ML Engineering

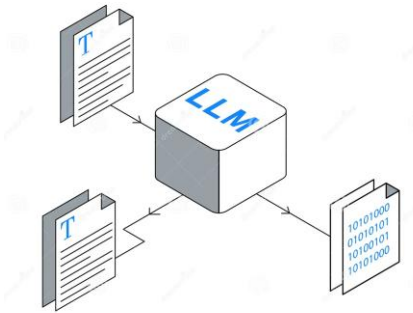
- Significant shift in focus moving from traditional Machine Learning (ML) Engineering towards AI Engineering
- While traditional ML engineering focuses on developing machine learning models *from scratch*, AI engineering builds on top of *existing models* to create intelligent applications



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Drivers leading to the rise of AI Engineering

- **Catalyst** - The emergence and widespread availability of powerful Foundation Models (FMs) and Large Language Models (LLMs)
- **Accessibility** - FMs and LLMs lower the barrier to entry, enabling more people to build AI applications without deep ML expertise
- **Product-Centric Approach** - AI Engineering often starts with *product development* and leveraging existing models, rather than data collection and model training



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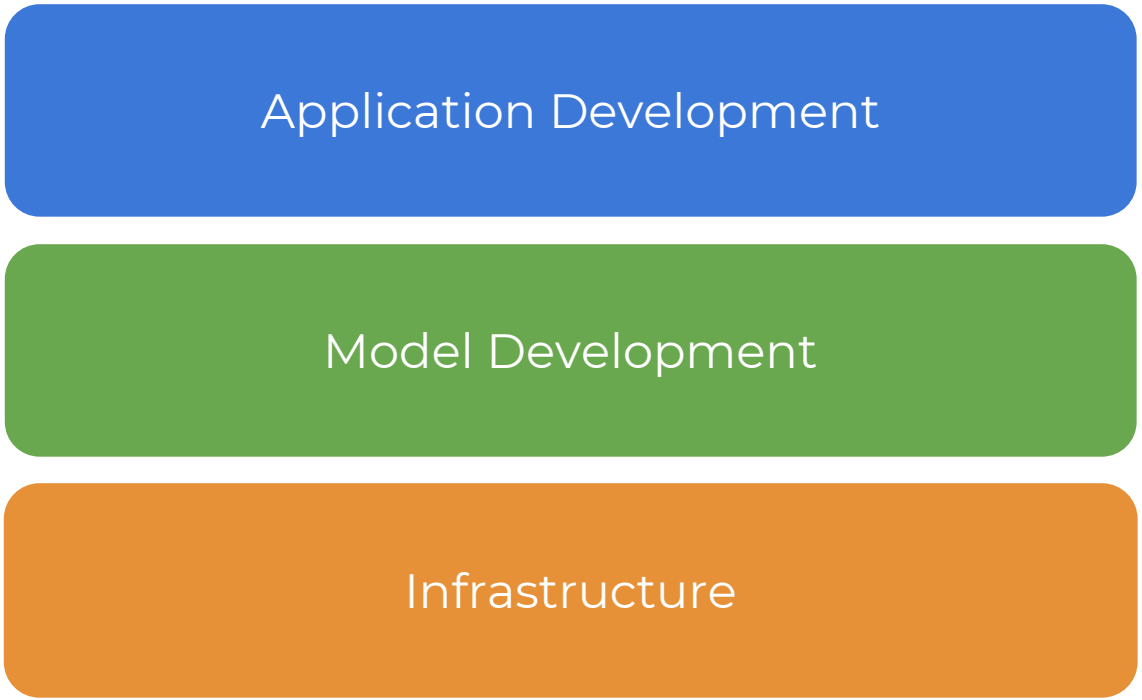
ML Engineering vs AI Engineering

	ML Engineering	AI Engineering
Model	Models are trained *from scratch*. Focuses more on *modeling and training*.	Uses foundation models. Focuses more on *model adaption*.
Compute Resources	Typically high during *training*.	Potentially lower due to *fine-tuning*.
Evaluation	Model outputs tend to be *close-ended* (e.g. classification). Easier to evaluate quantitatively.	Model outputs are *open-ended* (e.g. text generation). Harder to evaluate objectively and consistently.

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The AI Engineering Stack

Developing an AI application usually starts from the top layer and we move down the stack as needed



Focus

Prompt Crafting, Context Provision, Interface Design, User Experience

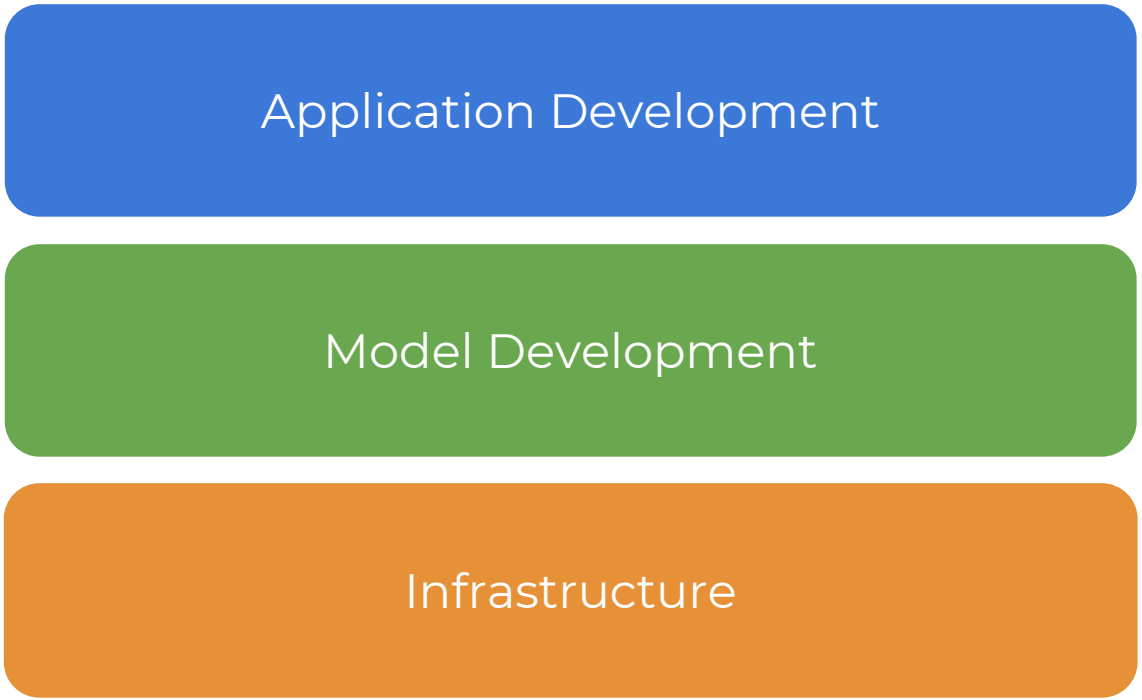
Adapting and fine-tuning models

Serving and deploying models efficiently

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The AI Engineering Stack

Understanding the AI Engineering Stack is key to grasping how modern AI applications are structured and developed.



- AI Interface
- Prompt Engineering
- Context Construction
- Evaluation

- Inference Optimization
- Dataset Engineering
- Modeling & Training
- Evaluation

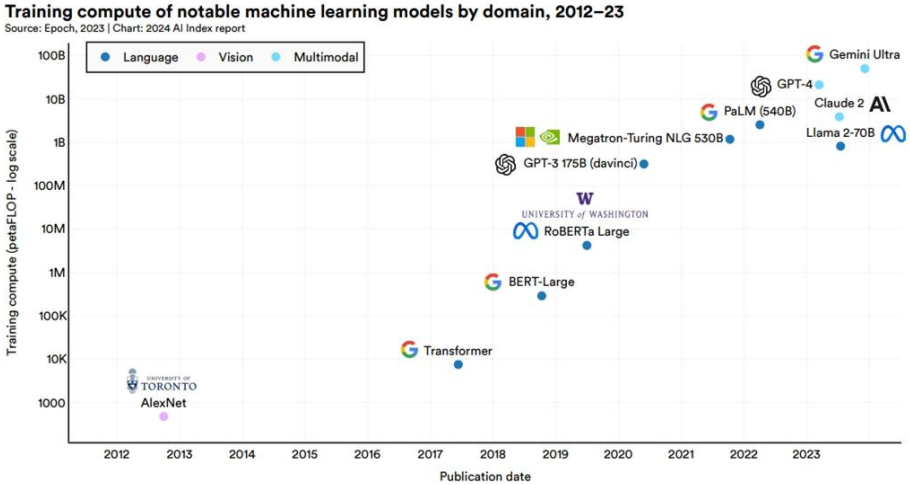
- Compute Management
- Data Management
- Serving
- Monitoring

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Foundation Models

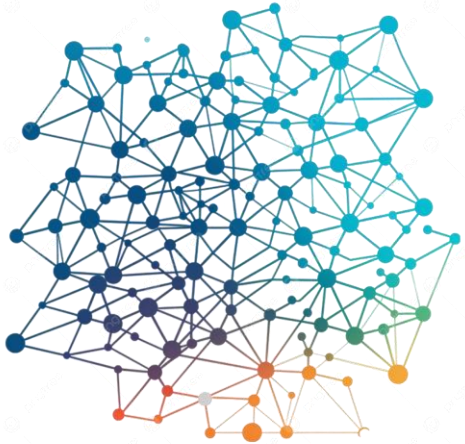
- Foundation models emerged from large language models
- Large scale, versatile AI models trained on a massive amount of diverse data
- Training a foundation model is an incredibly **complex** and **costly** process



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Evaluating Models

- Crucial and integral part of AI Engineering
- Mitigate risks and uncover opportunities
- Determine application usefulness and readiness for deployment
- Continuous improvement of models and applications



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Challenges of evaluating AI systems

- The challenges tackled by these models are often naturally complex
- The tasks they handle are usually **open-ended**, with multiple valid answers
- Foundation models often operate as **black boxes**, making their inner workings hard to interpret
- Public evaluation benchmarks tend to become "**saturated**" quickly - that is, models start achieving perfect scores as they improve
- For general-purpose models, it's important not only to evaluate existing tasks but also to uncover new, emerging capabilities

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Model Adaptation

- **Prompt-based techniques**
  - includes prompt engineering
  - adapting a model without updating the model's weights
  - provide instructions and context to the model instead of changing the model itself
- **Fine-tuning**
  - requires updating model's weights
  - more complicated and requires more data
  - improves model's quality, latency and cost significantly

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Prompt Engineering



- the process of crafting an instruction that gets a model to generate the desired outcome
- *no changes* to the model's internal parameters (weights)
- **prompt** - the instruction given to a model, can range from simple questions to complex tasks like researching competitors or analyzing data

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A prompt generally includes



- **Task description**: What you want the model to do, including the role you want the model to play and the output format
- **Example(s)** of how to do this task: Providing one or more examples of what the desired input and output should look like
- **The task**: The concrete task you want the model to do, such as the specific question to answer or item to process

Prompt:
You are a helpful assistant that rewrites sentences to be more polite.
When given a direct or blunt sentence, rewrite it in a way that sounds courteous and professional.

Example:
Input: "Send me the report now."
Output: "Could you please send me the report at your earliest convenience?"

Task:
Input: "You're wrong! Read the instructions again."
Output: [Your Turn]

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In-Context Learning



- teaching models what to do via *prompts*
- each example provided in the prompt is called a *shot*
- *zero-shot, few-shot learning*
- the number of examples needed depends on the model and the application

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When to use Prompt Engineering?



Use when

- we need **quick, low-cost customization**
- our use cases can be solved by **rephrasing the prompt** (e.g. summarization, translation, tone changes)
- the model performs well **out-of-the-box** with simple guidance

Pros: Fast, no infrastructure changes, no training needed

Cons: Limited in scope and consistency for complex tasks

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RAG (Retrieval-Augmented Generation)



- technique that enhances a model's generation by retrieving relevant information from **external memory sources**
- combines **document retrieval** with **language generation** to provide more accurate and up-to-date answers
- instead of relying only on what a language model was trained on, RAG first retrieves relevant information from external sources (like databases, vector stores, or knowledge bases) and then generates a response using that context

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RAG Architecture



- A basic RAG system consists of two main components: a **retriever** and a **generator**
- The process follows a **retrieve-then-generate pattern**
- First, a **retriever** component fetches relevant information from external data sources, such as internal databases, user chat sessions, or the internet
- Then, a **generator** component, typically a foundation model, uses this retrieved information, combined with the original user query, to produce a response

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How RAG works?



- **Embed documents** using an embedding model
- **Store those embeddings** in a vector database
- **Embed the user query** and search the vector database for similar chunks
- **Generate a response** using the retrieved documents as context

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When to use RAG?



Use when

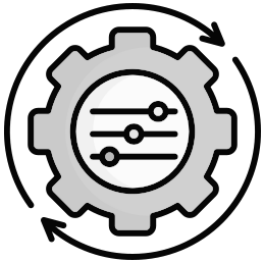
- we want the model to answer using **specific or up-to-date knowledge**
- the model's base knowledge is insufficient or **outdated**
- we need **contextual grounding** (e.g. using company documents, legal contracts, etc)

Pros: No re-training, better accuracy and transparency

Cons: Requires search infrastructure, may increase latency

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Fine-Tuning



Use when

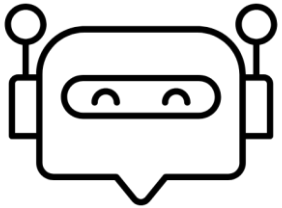
- we need **consistent behavior** on a **narrow, domain-specific** task
- prompt engineering and RAG are not good enough
- we have a **clean, labeled dataset** for training

Pros: Customized model behavior, can improve performance

Cons: Requires compute, time and maintenance

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Agentic AI (Tool-Using/Autonomous Agents)



Use when

- our use case involves **multi-step reasoning**, tool use (APIs, databases), or long tasks
- we want the AI to **plan, decide and act** (e.g. schedule a meeting, automate workflows)

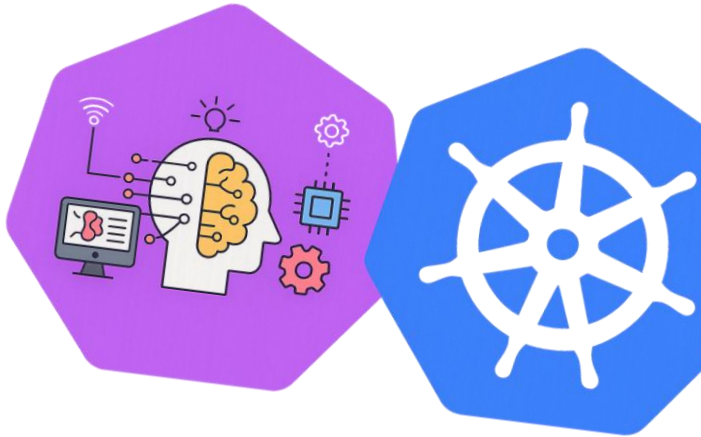
Pros: More flexible and powerful workflows

Cons: Harder to control, evaluate and troubleshoot. Requires orchestration

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Compute Management and Serving

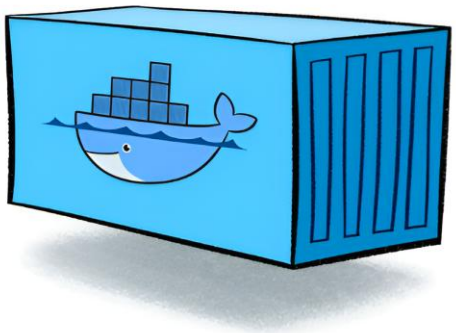
- Containers and Kubernetes play critical roles in **operationalizing AI engineering** by enabling scalable, reproducible, and efficient workflows
- We adopt **cloud-native** technologies and best practices, making it easier for AI engineers to:
  - use managed GPU services
  - optimize costs
  - adopt hybrid or edge AI deployments



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Reproducibility & Portability

- AI models require specific dependencies (libraries, versions, CUDA etc). Containers ensure that the environment is **exactly the same** across development, testing and production
- *AI engineers use containers to containerize models and workflows so they can run anywhere - on a laptop, in the cloud, or on a cluster - without worrying about environment issues*



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Scalability & Orchestration

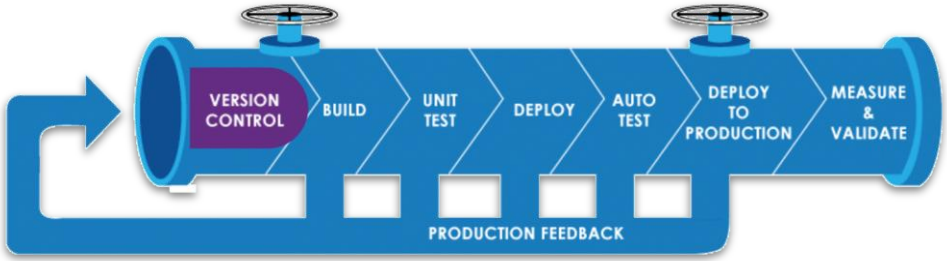
- AI workloads (training, RAG pipelines, agentic workflows or inferencing) need scalable infrastructure
- Kubernetes automates deployment, scaling, and management of these containers
- *Kubernetes allows AI engineers to scale model training across GPUs or deploy multiple inference endpoints reliably*



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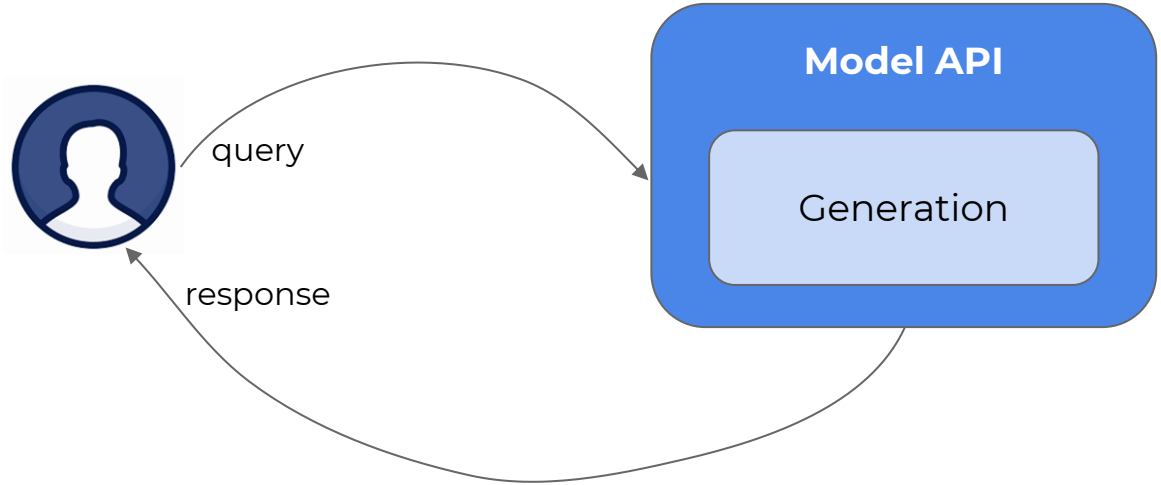
MLOps Integration for AI

- Containers + Kubernetes can be integrated into CI/CD pipelines for MLOps enabling
  - Continuous training
  - Automated model deployment
  - Monitoring and rollback
- *AI engineers bring DevOps best practices to AI systems, making them reliable and continuously improving*



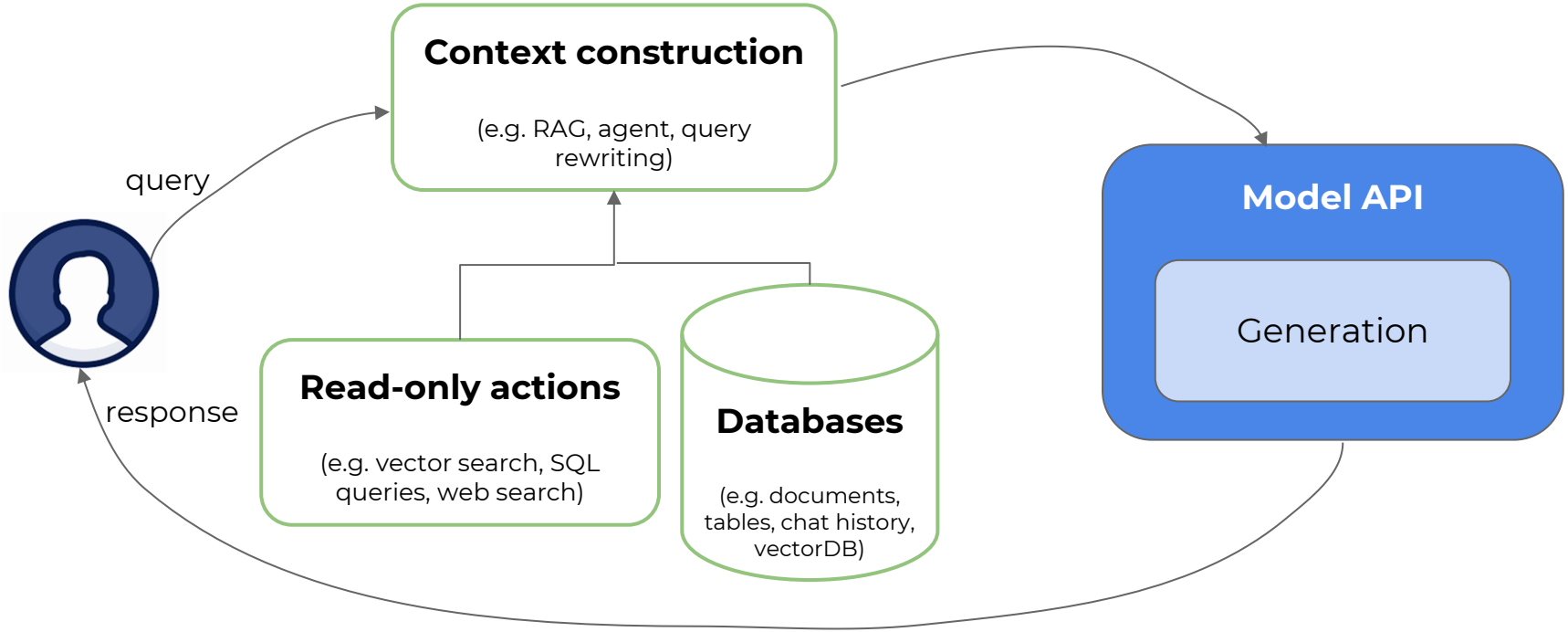
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AI Engineering Architecture



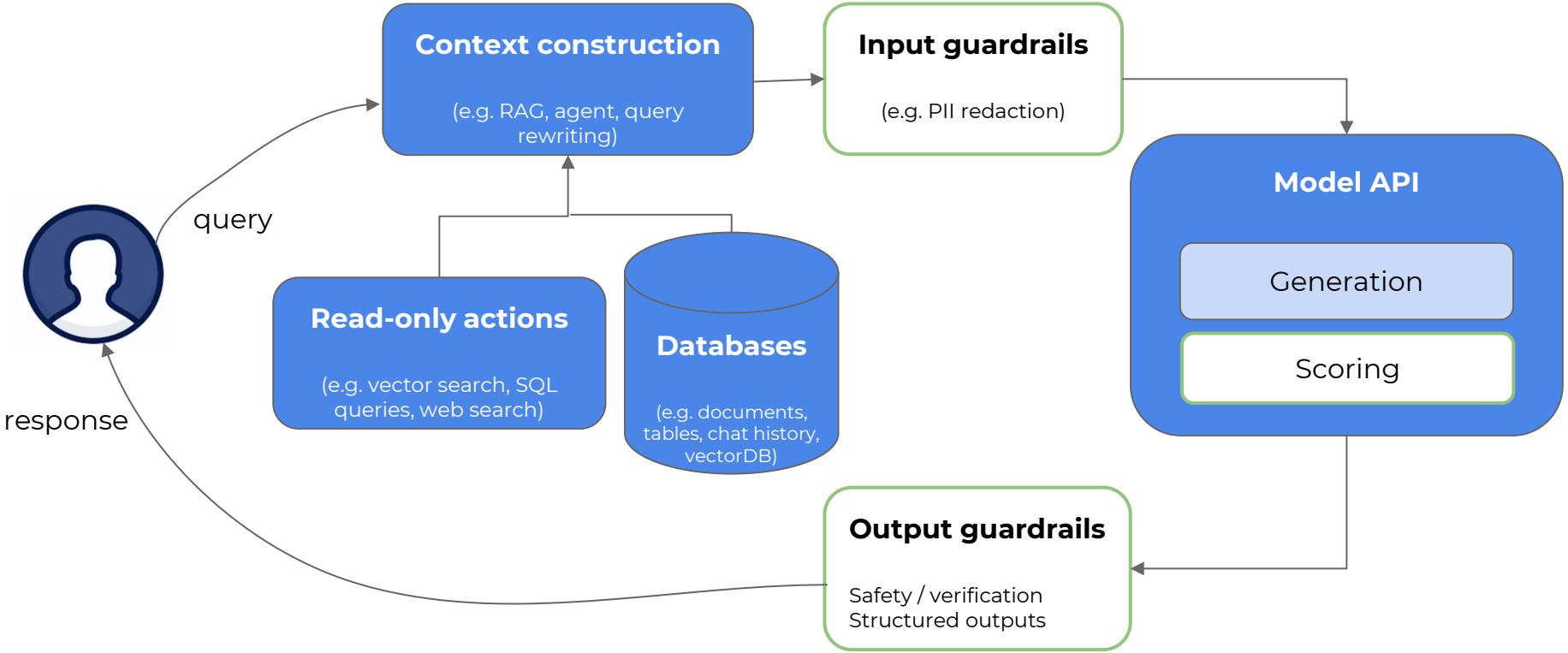
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AI Engineering Architecture



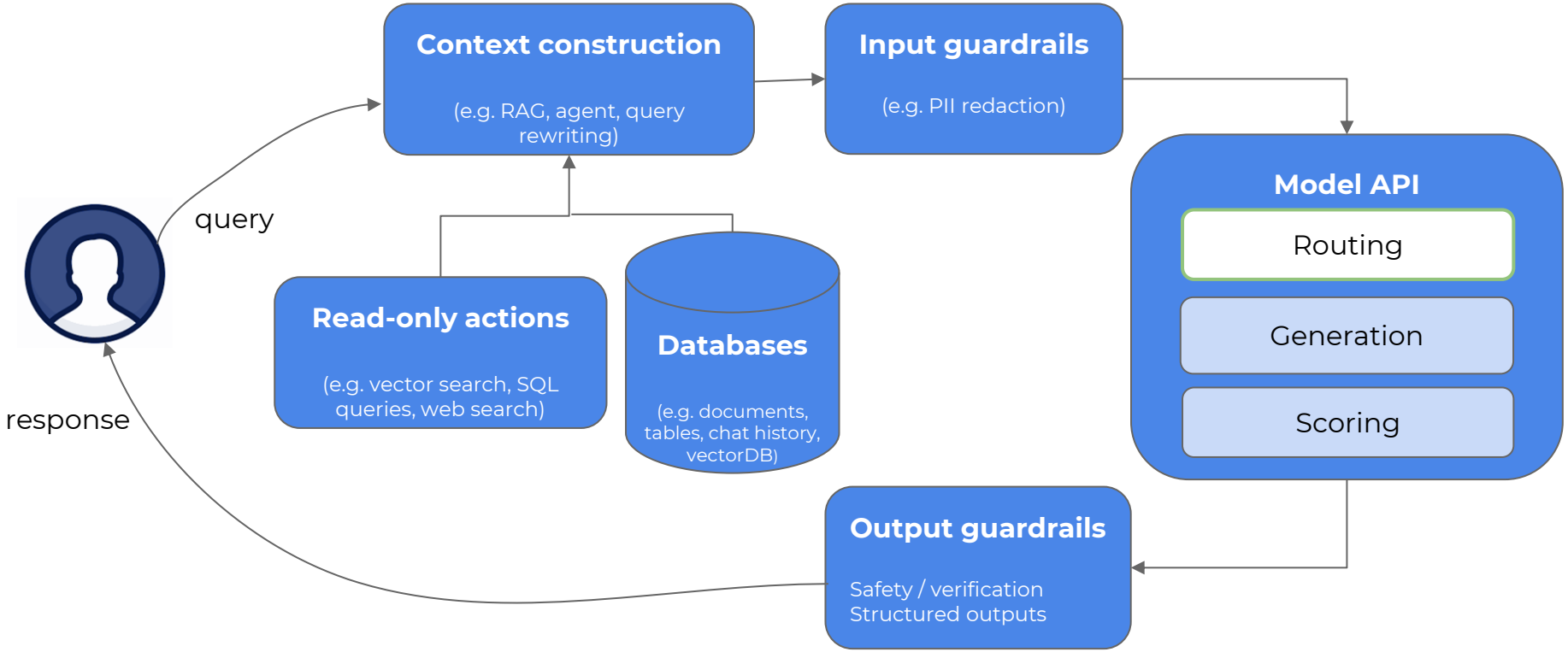
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AI Engineering Architecture



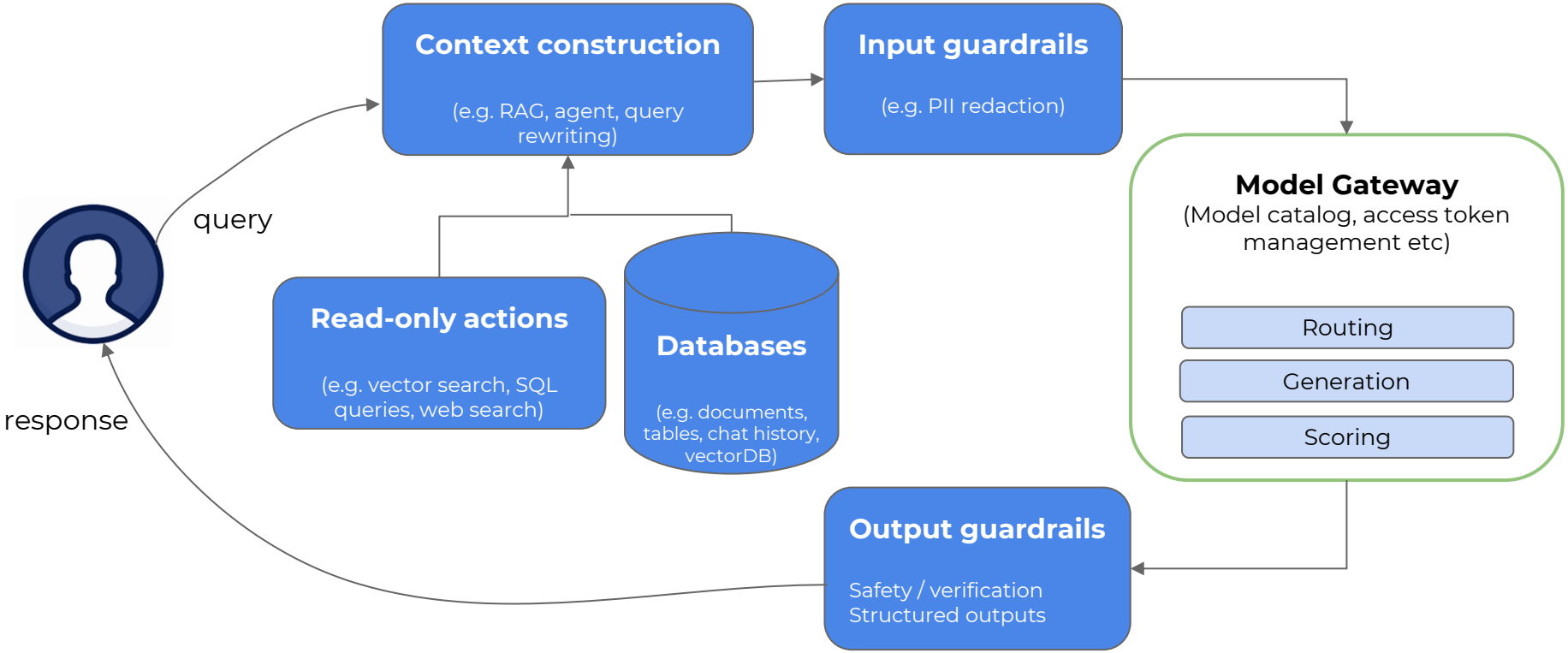
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AI Engineering Architecture



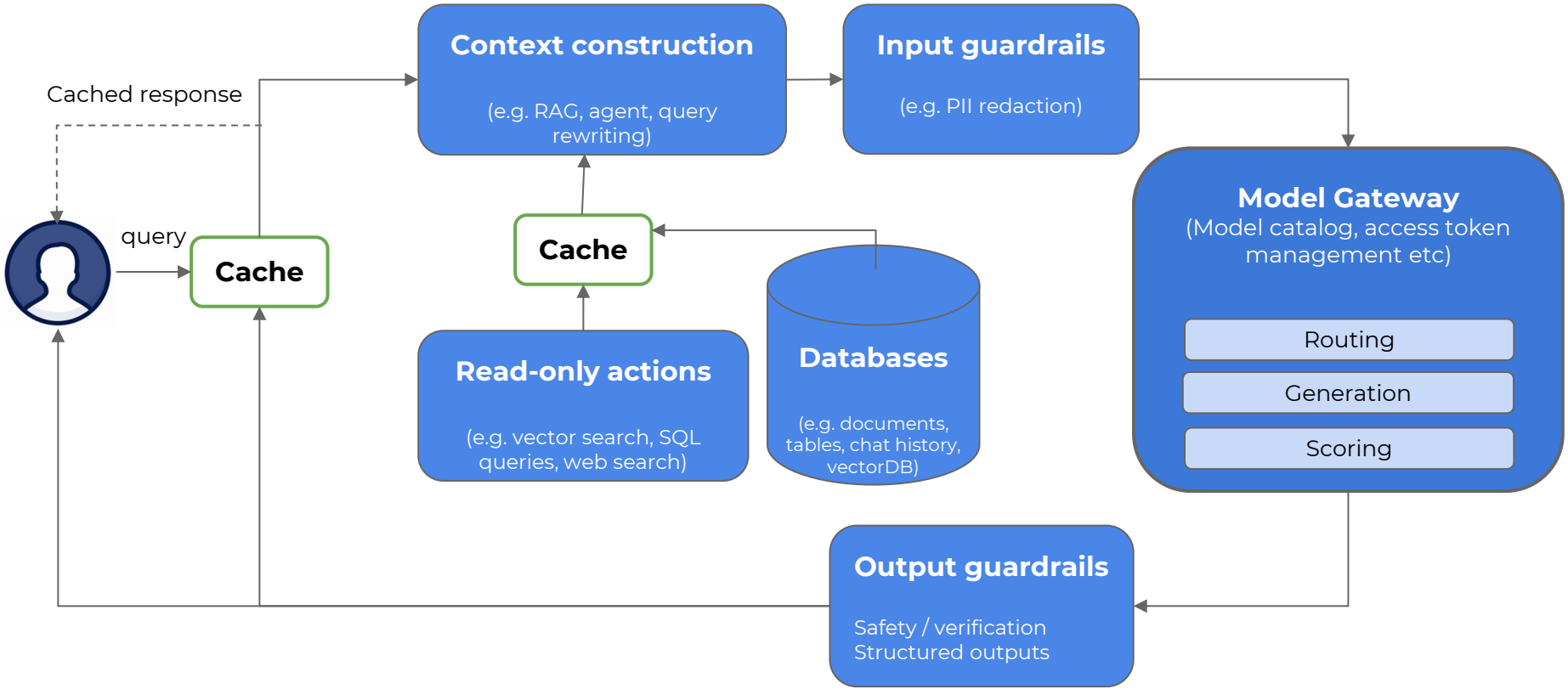
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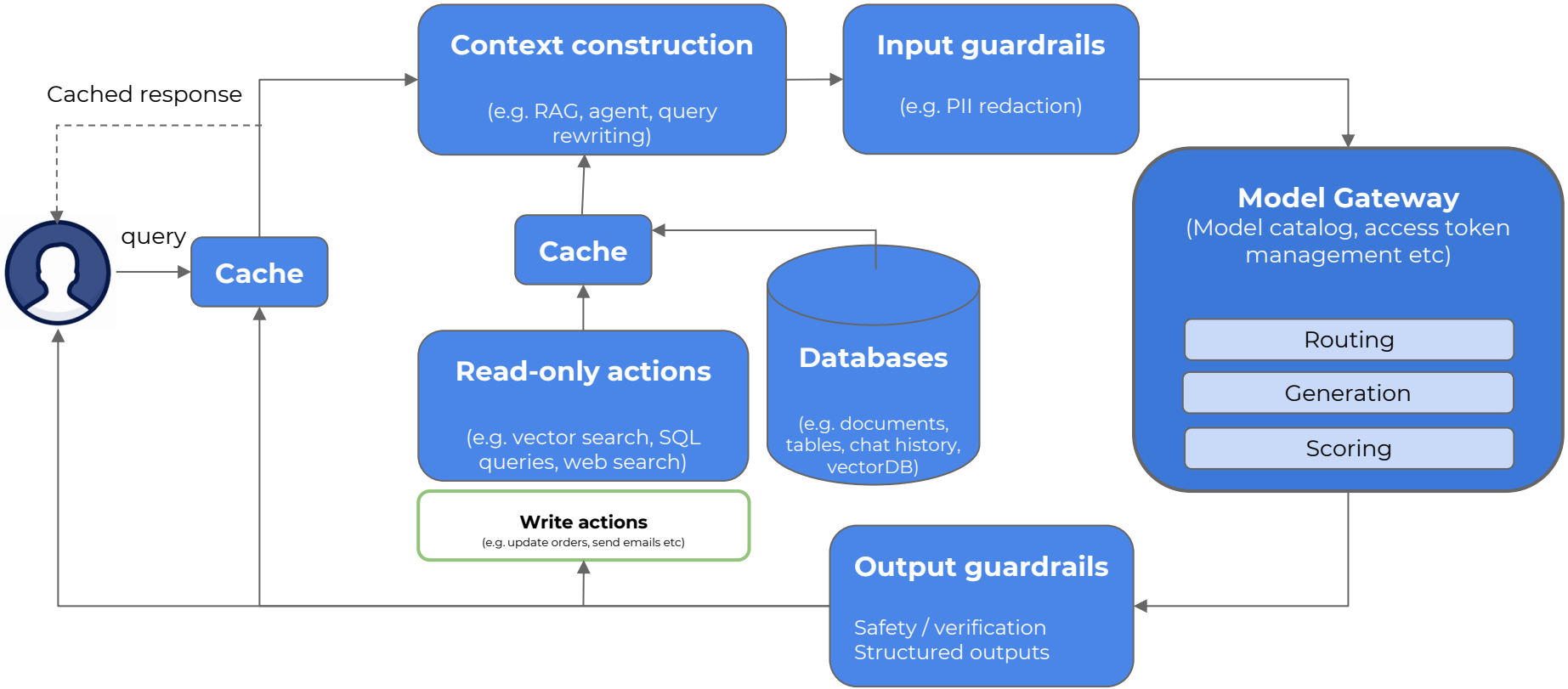
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AI Engineering Architecture



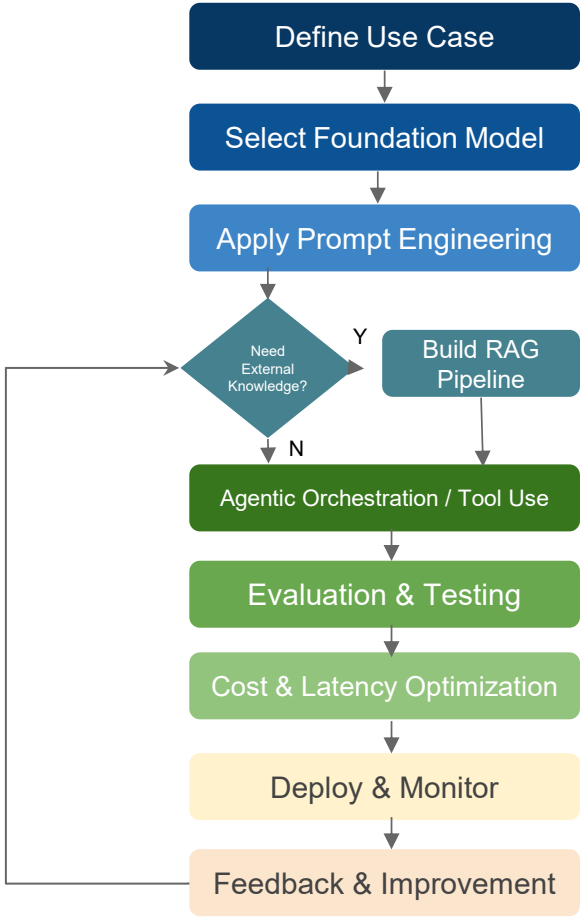
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AI Engineering Architecture



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Workflow



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Key Themes of AI Engineering

- **Foundation Models and Evaluation** - AI engineering differs from traditional ML engineering by focusing less on training models and more on prompting, adapting, integrating, and deploying large-scale models via APIs
- **Model Adaptation Techniques** - When to use Prompt engineering, Retrieval-Augmented Generation (RAG), Fine-tuning, Agents etc
- **Deployment: Scalability and Optimization**
    - Real-world strategies for reducing inference latency, managing costs (via quantization, pruning, distillation), and choosing infrastructure like GPUs/TPUs
    - A broader product-driven mindset: advising engineers on business strategy and user-centered design

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Dell AI Engineering Fundamentals



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